

Junye Zhou

Portfolio: moonlitjaderabbit.github.io | junye.zhou@mail.utoronto.ca

EDUCATION

University of Toronto

Toronto, ON

Bachelor of Applied Science in Computer Engineering

Expected Graduation: 2030

- Current coursework: Digital Systems (Verilog), Programming Fundamentals (C++), Intro to Electronics, Advanced Engineering Mathematics
- Completed coursework: Electrical Circuits, Algorithms & Data Structures (Python, C), Structural Analysis, Linear Algebra, Engineering Mathematics & Computation

TECHNICAL SKILLS

Languages: C, Python, Java, C++, Verilog

Tools: MATLAB, Quartus Prime, LTspice, OSP Tracker

Engineering: Requirements Engineering, Pugh Matrices, Concept Selection, Verification & Testing, Proxy Testing

PROJECTS

Variable Resistance Swim Training System [↗](#)

Feb 2026 – Apr 2026

- Developed a variable-resistance swim training system for the Mississauga Aquatic Club (MSSAC) in response to a client RFP
- Designed a capstan-based mechanism providing **8 adjustable resistance levels (60–170 N)** with approximately **3-second switching time**
- Evaluated alternative resistance mechanisms using trade-off analysis and Pugh matrices, balancing adjustability, manufacturability, and poolside safety
- Created engineering requirements and verification plans; presented the final design at the Praxis II Showcase

Matboard Bridge [↗](#)

Nov 2025 – Jan 2026

- Ranked **#1 in cohort** and **#2 in EngSci Class of 2025** for bridge design and performance
- Designed, analyzed, and fabricated a 1.25 m matboard bridge with a predicted failure load of **954 N**; full-scale testing reached approximately **1000 N**
- Performed hand calculations and compared results with team-developed Python structural analysis [↗](#) to verify predicted loads and failure modes
- Iterated through four design revisions involving diaphragm placement and splice reinforcement while meeting strict material constraints

Damped Pendulum Lab [↗](#)

Oct 2025 – Jan 2026

- Recorded pendulum motion at **60 fps** and performed frame-by-frame motion tracking using OSP Tracker
- Analyzed period-angle relationships, damping, length scaling, and quality factor, obtaining $Q \approx 109$ using two independent methods
- Quantified experimental uncertainty and compared measured behavior against theoretical predictions

Alpha Release Project [↗](#)

Sept 2025 – Dec 2025

- Designed a solution addressing conflicting sleep and study lighting requirements in shared residence rooms
- Generated four design concepts and evaluated them using Pugh matrices and illuminance proxy testing in a real residence room
- Recommended a foldable bed-dome concept targeting < 1 **lux** at the sleeper's eyes

ADDITIONAL EXPERIENCE

Soma Bone Broth

Toronto, ON

Kitchen Hand

Jan 2022 – Feb 2022

- Supported food preparation and workstation organization in a fast-paced commercial kitchen